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Multi-agent SEA: step-wise evidence acquisition for multimodal financial reasoning

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ABSTRACT

Multimodal financial reasoning requires integrating heterogeneous information sources, including textual context, visual signals, and domain-specific knowledge. Despite recent advances, multimodal large language models (MLLMs) still suffer from perceptual errors and insufficient knowledge, leading to incorrect intermediate evidence and, ultimately, unreliable reasoning outcomes. We propose SEA (Step-wise Evidence Acquisition), a multi-agent framework that models multimodal financial reasoning as an iterative process of task decomposition, evidence acquisition, and reasoning. The multi-agent SEA introduces a planner agent that decomposes a complex question into a sequence of subtasks and generates natural-language queries under a unified communication protocol. These queries are routed to specialized evidence-seeking agents, which retrieve complementary evidence from visual inputs, textual context, and external financial knowledge. The retrieved information is accumulated into a dynamically evolving evidence state, enabling the reasoning agent to perform step-wise, evidence-grounded inference. Extensive experiments on the financial multimodal benchmarks demonstrate that SEA significantly improves reasoning accuracy and robustness compared to strong baselines. Further analysis shows that our framework effectively alleviates perception errors and enhances reasoning performance.

1. Introduction

Recent advancements in multimodal large language models (MLLMs) have significantly improved AI systems' capacity to integrate visual and textual information [1–3]. These models have demonstrated strong performance across a wide range of tasks [4], including visual question answering [5], and multimodal reasoning [6]. However, their performance on multimodal financial reasoning and question-answering tasks remains far from satisfactory [7–9]. Financial problems often require precise interpretation of numerical information, domain-specific knowledge, and multi-step reasoning over heterogeneous inputs such as text, tables, and financial charts [10,11]. These characteristics pose significant challenges for current multimodal models.

The latest financial benchmarks, e.g., FinMMR [12], FAMMA [7], and FinMME [8], find that state-of-the-art MLLMs consistently fail in reliable financial reasoning. These failures are primarily attributed to perception errors (misreading charts), context misunderstanding (misinterpreting constraints), and knowledge insufficiency (lacking domain formulas), as illustrated in Fig. 1. While diverse in origin, these issues

converge on a single critical failure: the inability to acquire the correct intermediate evidence. Consequently, models proceed with reasoning based on incomplete or irrelevant information, leading to cascading errors.

In contrast, human expert reasoning follows a distinct, decomposed pattern [13]. Aligned with information processing theory [14] and bounded rationality [15], experts adopt a step-wise strategy to manage limited cognitive resources. Rather than simultaneous processing, they employ goal-directed attention [16] to identify and extract task-relevant evidence from heterogeneous sources dynamically. In this framework, each reasoning step determines the specific evidence required for the next, making accurate evidence acquisition the bottleneck for successful inference. This suggests that financial reasoning performance is limited less by raw logic than by the ability to retrieve precise, task-conditioned information during intermediate steps.

This observation raises a fundamental question: *Is multimodal financial reasoning fundamentally limited by reasoning capability, or by the ability to acquire accurate and task-relevant evidence during intermediate steps?* We hypothesize that the primary source of reasoning errors lies in incorrect

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1566-2535/© 2026 Elsevier B.V. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Context (c): Financial analysts often assess the value of fixed-income securities when comparing different investment opportunities. Estimating the present value of future payments helps determine whether a bond offers a reasonable return.

Question (q): A government bond has a face value of \$1,000 which matures in 5 years. Given an interest rate of 4%, what is the present value of this bond?

Image (I):



Options (O): A: \$1133.56, B: \$1148.65, C: \$1159.27, D: \$1142.15

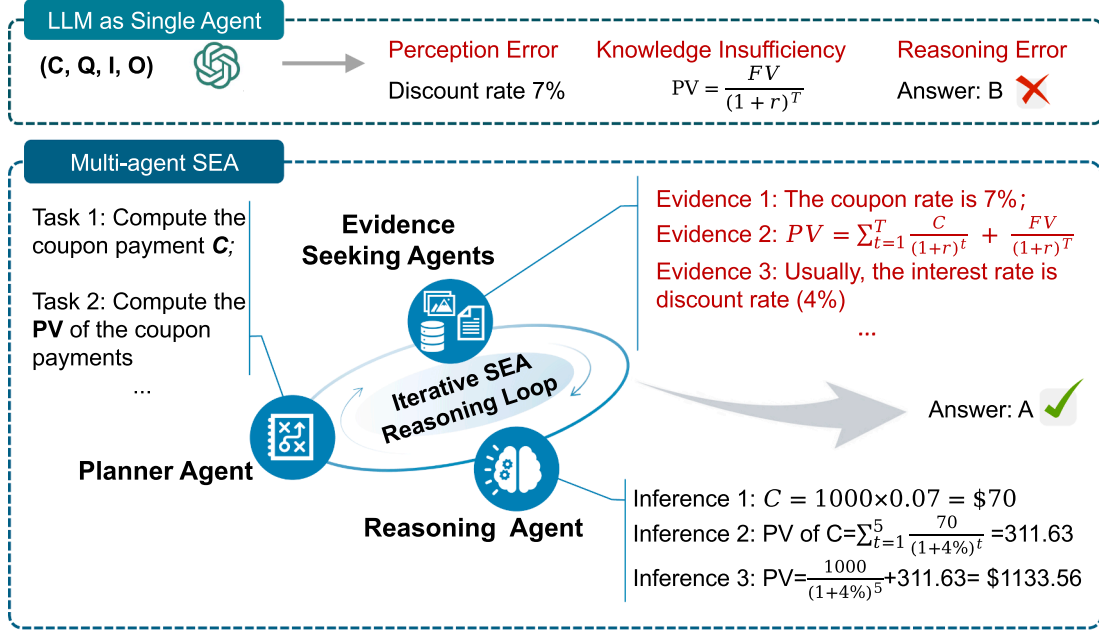


Fig. 1. An illustrative example of financial multimodal reasoning via single MLLM and SEA.

or incomplete intermediate evidence, rather than deficiencies in reasoning itself.

In this work, we are motivated to propose SEA (Step-wise Evidence Acquisition) for multi-agent systems to address the challenge of cascading reasoning failures caused by poor intermediate evidence quality. SEA explicitly models multimodal financial reasoning as a step-wise process of evidence acquisition and reasoning. Instead of forcing a single model to process all multimodal inputs in a single pass, SEA decomposes the reasoning process into a sequence of sub-tasks, where each step identifies and retrieves the specific evidence required for subsequent inference. Specifically, a planning agent first decomposes the task into sub-tasks and determines the information needed at each step. It then generates targeted evidence-seeking queries, which are routed to specialized agents responsible for extracting information from different sources, including financial images, textual context, and external financial knowledge. The retrieved evidence is accumulated in a shared evidence memory and used to guide step-wise reasoning. To support collaboration among agents, SEA adopts a query-based communication mechanism that enables targeted evidence exchange among agents. Rather than relying on complex structured protocols, the planner issues concise natural-language queries, and specialized agents return focused responses. This design mirrors how human experts solve financial problems by asking targeted questions to obtain necessary information, re-

ducing redundant information, and improving the quality of intermediate evidence.

We evaluate our proposed multi-agent SEA on three benchmarks: FinMR, FAMMA, MMMU-Finance [7,9,17]. The experimental results demonstrate its strong performance over the state-of-the-art baselines, yielding average gains by 13.68% accuracy. Furthermore, the suggested dynamic task decomposition strategy yields 6.14% accuracy gains on average, compared to the static strategy. Finally, the query-based communication mechanism exceeds previous methods, e.g., pipeline-based [18], controller-based [19], and debate-based [20] mechanisms, by at least 4.17%, indicating that our query-based methods can better align the collaboration between multi-agents, deliver evidence with higher utilities for the downstream reasoning.

Our contributions can be summarized as follows:

- We propose SEA, a multi-agent framework that explicitly models task decomposition, evidence acquisition, and evidence-guided reasoning, enabling dynamic integration of visual, textual, and knowledge-based information.
- We introduce a query-based communication mechanism in which agents interact via concise, step-conditioned natural-language queries, enabling effective and interpretable evidence exchange.
- Extensive experiments on three multimodal financial reasoning benchmarks demonstrate that SEA consistently outperforms strong

baselines and mitigates perception errors and knowledge insufficiency by improving intermediate evidence quality.

The remainder of this paper is organized as follows. Section 2 reviews related work on financial multimodal reasoning, active information acquisition, and multi-agent/tool-augmented reasoning, positioning SEA within existing studies. Section 3 presents the proposed SEA framework, including the problem formulation, planner-guided task decomposition, multi-source evidence acquisition, evidence-guided reasoning, and query-based communication protocol. Section 4 describes the experimental setup, benchmark datasets, and baseline methods used for evaluation. Section 5 reports the main experimental results and provides further analyses of dynamic sub-task generation, evidence acquisition quality, agent communication strategies, and remaining error patterns. Finally, Section 6 concludes the paper and discusses future research directions.

2. Related work

2.1. Financial multimodal reasoning

In the financial sector, characterized by intricate decision-making processes, domain-specific expertise, and sophisticated analysis, LLMs have demonstrated substantial potential in enhancing financial reasoning and analytical tasks, such as investment decision-making [21] and market sentiment analysis [4,22,23]. Recent evaluations demonstrate strong performance of LLMs on textual financial reasoning datasets, including TAT-QA [24], and FinQA [10]. However, real-world financial reasoning rarely relies on a single textual source. Rather, it typically requires the integration of heterogeneous information from multiple sources.

Generally, financial multimodal reasoning tasks can be categorized into two types: *knowledge-driven reasoning* and *perception-driven reasoning*. Knowledge-driven reasoning [7,9,25] requires MLLMs to leverage deep financial expertise, handle advanced financial terminologies, and interpret nuanced contextual information. In contrast, perception-driven reasoning [8] mainly assesses visual capabilities, requiring interpretation of complex financial visual data such as charts, graphs, and tables. Despite MLLMs achieving strong performance on multimodal tasks across many domains [26], their performance on financial multimodal benchmarks, such as FinMR [9], FAMMA [7], and FinMME [8], remains weak. Knowledge-driven reasoning is limited by weak command of financial expertise and implicit professional conventions. Perception-driven reasoning is hindered by inaccurate extraction of fine-grained information from dense charts and tables. As a result, MLLMs often miss critical knowledge, numerical details, and visual cues needed for correct financial reasoning.

Beyond benchmark-oriented financial reasoning, recent work has also explored LLM agents for financial decision-making, especially in trading scenarios. TradingAgents simulates a real-world trading firm through multiple specialized agents, including fundamental analysts, sentiment analysts, technical analysts, traders, bull and bear researchers, and risk managers [27]. A recent survey further summarizes the architectures, data inputs, backtesting protocols, and challenges of LLM-based financial trading agents [28]. These works demonstrate the potential of agent-based financial decision systems in the financial domain. However, they mainly focus on market trading and portfolio actions, whereas our work studies multimodal financial reasoning over charts, tables, textual contexts, and domain knowledge.

2.2. Active information acquisition

Sequential information acquisition has been widely studied as a way to reduce unnecessary computation and improve decision efficiency. Early works in adaptive computation introduced dynamic inference mechanisms, such as adaptive computation time [29], which allow

neural networks to learn halting decisions during reasoning. Related approaches include early-exit Transformers [30] and selective attention mechanisms that dynamically allocate computation resources. Beyond efficiency-driven adaptive models, active feature acquisition has been studied under a decision-theoretic framework. Classification with costly features was formulated as a reinforcement learning problem, where the agent sequentially decides which features to observe [31]. Similarly, sequential feature acquisition under uncertainty was proposed to balance accuracy and acquisition cost [32]. These methods explicitly model information cost but focus primarily on tabular or static feature settings. In vision and multimodal domains, active perception has been explored through glimpse-based models [33] and sequential attention mechanisms that learn where to look next [34]. More recently, adaptive token selection strategies in Transformers [35] dynamically prune irrelevant tokens to improve efficiency.

Recent studies on MLLMs also highlight the importance of selectively using visual information. Visual In-Context Learning improves MLLMs by retrieving visual demonstrations, summarizing images according to task intent, and composing intent-oriented demonstrations for downstream reasoning [36]. This line of work suggests that multimodal reasoning benefits from task-relevant visual information selection rather than indiscriminate visual encoding.

However, existing approaches typically optimize for computational efficiency, or feature cost, or demonstration selection, rather than the utilities of structured reasoning. In contrast, our work frames multimodal financial reasoning as a sequential information acquisition process in which the model explicitly determines what evidence is needed from images, textual context, and an external financial knowledge base, as well as when to stop. Instead of passively encoding all modalities, our system actively controls when and how evidence is retrieved, enabling demand-driven reasoning.

2.3. Multi-agent and tool-augmented reasoning

Recent advancements in LLMs have shown that integrating external tools or coordinating multiple agents can significantly enhance reasoning capabilities [37,38]. Tool-augmented frameworks such as ReAct [37] interleave reasoning and acting, enabling models to query external environments during inference. Similarly, ToolFormer [38] equips language models with the ability to decide when to invoke external tools within a generation process. These approaches extend language models beyond passive text generation, allowing interaction with structured resources such as calculators, search engines, and code execution environments.

Retrieval-augmented reasoning further improves the factuality and reliability of complex reasoning by incorporating external knowledge into intermediate reasoning steps. For example, RARE extends a mutual reasoning framework with retrieval actions and a retrieval-augmented factuality scorer to prioritize factually grounded reasoning paths [39]. Related studies on multilingual knowledge graphs also demonstrate that structured knowledge, relational semantics, and contextual enrichment are important for knowledge-intensive reasoning and alignment tasks [40,41]. These studies suggest that external knowledge sources can provide useful semantic and relational evidence for complex reasoning.

In parallel, multi-agent reasoning systems decompose complex tasks into specialized roles, such as planner, executor, and verifier agents. Frameworks such as AutoGen [19] and subsequent works [42] coordinate multiple LLM-based agents through iterative communication, enabling collaborative problem solving across multiple steps. Recent studies have also extended role-specialized collaboration to high-stakes multimodal domains. For instance, MAM decomposes multimodal medical diagnosis into specialized agents, including a general practitioner, specialist team, radiologist, medical assistant, and director, demonstrating the effectiveness of modular collaboration across text, image, audio, and video modalities [43]. These systems typically rely on prompt-based

Table 1

Comparison of representative techniques, including their main ideas, advantages, and limitations.

Technique	Representative Works	Main Idea	Advantages	Limitations
Multimodal CoT	FinMR [9]; FAMMA [7]; FinMME [8]; FinMMR [12]	Evaluates models on financial questions involving textual context, charts, tables, and domain knowledge.	Provides realistic benchmarks for assessing financial expertise, numerical reasoning, and visual understanding.	Existing MLLMs often suffer from perception errors, weak domain knowledge, and insufficient cross-modal alignment.
LLM-based financial agents	TradingAgents [27]; LLM trading agent survey [28]	Uses multiple LLM agents to simulate financial decision-making workflows such as market analysis, trading, and risk assessment.	Demonstrates the potential of role-specialized agents in financial decision systems.	Mainly focuses on trading or portfolio actions rather than evidence-grounded multimodal financial reasoning.
Active information acquisition	Adaptive computation time [29]; active feature acquisition [31,32]; active perception [33,34]; adaptive token selection [35]	Dynamically selects features, tokens, or visual regions to reduce unnecessary computation or acquisition cost.	Improves computational efficiency and encourages selective information processing.	Usually optimizes efficiency or feature cost, rather than task-conditioned evidence utility for multi-step reasoning.
Visual information selection	Visual In-Context Learning [36]	Selects and summarizes task-relevant visual demonstrations or visual information for MLLM reasoning.	Shows that selective visual evidence can improve multimodal reasoning.	Focuses on demonstration selection or visual prompting, without explicit step-wise evidence acquisition.
Tool-augmented reasoning	ReAct [37]; ToolFormer [38]	Enables LLMs to interact with external tools such as search engines, calculators, or code executors.	Extends LLMs beyond passive generation and supports external knowledge or computation.	Tool use is often implicit or loosely controlled, and may not retrieve the precise evidence needed at each reasoning step.
Retrieval-augmented reasoning and KG-based methods	RARE [39]; RSA [40]; AKR [41]	Incorporates external knowledge, retrieval actions, or structured relational semantics into reasoning.	Improves factual grounding and highlights the value of structured knowledge and contextual enrichment.	Usually focuses on knowledge retrieval, factuality, or entity alignment, rather than multimodal financial evidence acquisition.
Multi-agent reasoning	AutoGen [19]; MAM [43]; role-aware agents [18]; debate-based agents [20]	Decomposes complex tasks into multiple specialized agents that collaborate through predefined roles or communication protocols.	Supports modular problem solving and role-specialized reasoning.	Existing collaboration is often prompt-driven, static, or debate-oriented, which may introduce redundant or misaligned evidence.
SEA (ours)	This work	Performs step-wise task decomposition, targeted evidence acquisition, and evidence-guided reasoning through query-based multi-agent communication.	Dynamically retrieves task-relevant evidence from visual inputs, textual context, and financial knowledge, improving reasoning reliability and interpretability.	Still depends on the quality of planner-generated queries and evidence returned by specialized agents.

role assignment and predefined communication protocols to guide agent interactions.

Despite their effectiveness, most existing approaches remain largely prompt-driven and are designed for general-purpose reasoning, trading decisions, retrieval-augmented knowledge reasoning, or domain-specific diagnosis. They do not explicitly address the challenges of multimodal financial reasoning, where accurate evidence acquisition from heterogeneous sources, including textual context, tables, charts, and financial knowledge, is critical. Moreover, existing frameworks often lack mechanisms for targeted information seeking, leading to redundant or misaligned evidence that propagates errors in downstream reasoning.

In contrast, our approach introduces a structured framework for evidence acquisition tailored to multimodal financial reasoning. Rather than relying on implicit tool usage, static demonstrations, retrieval alone, or loosely coordinated agent interactions, we design a simple yet effective communication protocol that enables agents to exchange concise, task-driven queries and responses iteratively. By focusing on targeted information acquisition and agent communication, our framework improves both the relevance of evidence and the reliability of downstream reasoning without requiring additional model training. Table 1 provides a comparison of representative techniques we discussed, including their main ideas, advantages, and limitations.

3. Methodology

3.1. Problem formulation

We study multimodal financial reasoning as a sequential evidence-grounded decision problem over heterogeneous inputs. Each instance is

defined as

$$X = (C, I, Q, O), \quad (1)$$

where C denotes the textual context (e.g., the context of a question), $I = \{i_1, \dots, i_S\}$ denotes the set of visual inputs (e.g., charts, tables, or figures), Q is the question, and $O = \{o_1, \dots, o_M\}$ is the set of candidate answers (options). The ground-truth answer is $Y^* \in O$.

Unlike conventional single-pass prediction, we model reasoning as an iterative process that alternates between *planning*, *evidence acquisition*, and *evidence-guided reasoning*. At reasoning step n , the system maintains: an evidence state E_n , which stores all retrieved evidence up to step n ; and a reasoning trajectory R_n , which stores all intermediate reasoning outputs up to step n . Formally,

$$E_n = \{e_1, \dots, e_{|E_n|}\}, \quad R_n = \{r_1, \dots, r_n\}. \quad (2)$$

The process starts from empty memory: $E_0 = \emptyset, R_0 = \emptyset$.

The goal is to predict the final answer $\hat{Y} \in O$ by progressively collecting task-relevant evidence and conditioning each reasoning step on the accumulated evidence. This formulation explicitly separates *what information is needed* from *how the answer is inferred*, thereby reducing hallucination and improving interpretability. For clarity, the main notations used in the subsequent formulation and analysis are summarized in Table A.8 in Appendix A.

3.2. Overview of SEA

SEA (Step-wise Evidence Acquisition) is a multi-agent framework that solves multimodal financial reasoning through a closed-loop interaction among three functional modules:

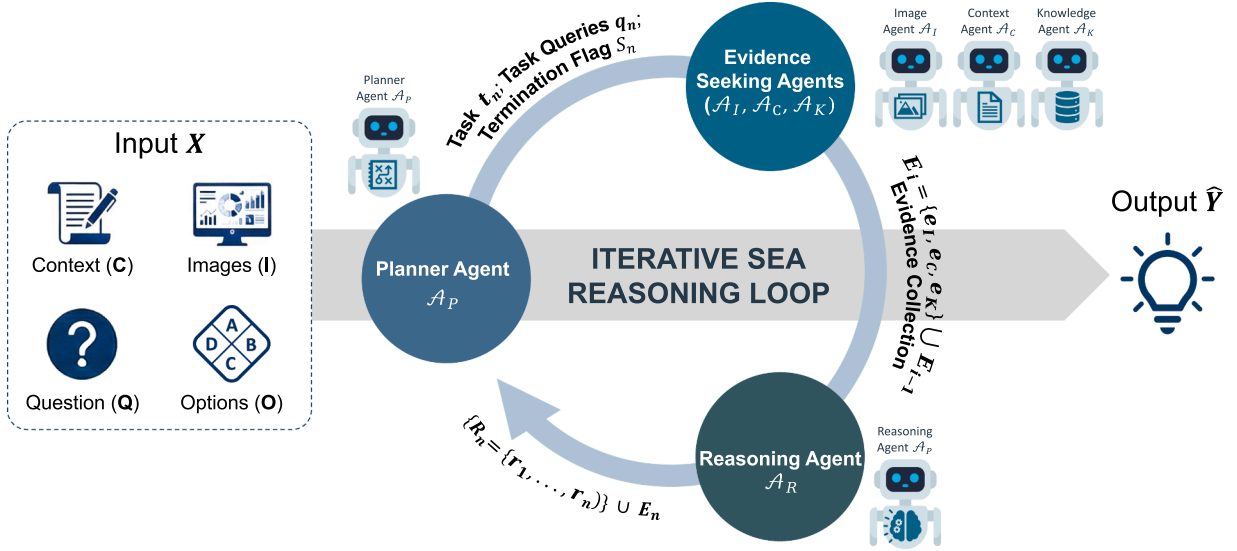


Fig. 2. Overview of SEA, a multi-agent framework for step-wise evidence acquisition. The planner generates sub-task-conditioned queries under a unified communication protocol, which are routed to multi-source evidence agents (image, text, and knowledge). Retrieved evidence is accumulated into a shared evidence state and used by the reasoning agent for step-wise inference. The entire process forms a closed-loop interaction that tightly couples planning, evidence acquisition, and reasoning.

1. a planner agent $\mathcal{A}_P$ that decomposes the problem into step-wise sub-tasks and issues evidence-seeking queries;
2. a set of evidence-seeking agents $\{\mathcal{A}_I, \mathcal{A}_C, \mathcal{A}_K\}$ that retrieve evidence from visual inputs, textual context, and external financial knowledge, respectively;
3. a reasoning agent $\mathcal{A}_R$ that produces intermediate reasoning conditioned on the retrieved evidence including report issues of sufficient evidence.

As illustrated in Fig. 2, SEA proceeds iteratively. At each step, the planner determines the current sub-task, formulates a set of targeted natural-language queries, and decides whether the current evidence is sufficient to stop. The queries are routed to the appropriate evidence-seeking agents, whose outputs are integrated into the shared evidence state. The reasoning agent then updates the reasoning trajectory based on the current sub-task and accumulated evidence. The process repeats until the planner determines that the available evidence is sufficient to answer the original question.

This design explicitly treats multimodal financial reasoning as a process of *demand-driven evidence acquisition*: the system does not attempt to process all information uniformly at once, but instead retrieves the specific evidence required by the current reasoning step.

3.3. Planner-guided task decomposition

The planner agent $\mathcal{A}_P$ controls the overall reasoning process. At step n , given the input instance X , the previous evidence state E_{n-1} , and the previous reasoning trajectory R_{n-1} , the planner outputs a sub-task t_n describing the current reasoning objective; a set of natural-language queries $q_n = \{q_{n,1}, \dots, q_{n,L_n}\}$ for evidence acquisition; and a termination flag $s_n \in \{\text{continue}, \text{stop}\}$. Formally,

$$(t_n, q_n, s_n) = \mathcal{A}_P(X, E_{n-1}, R_{n-1}). \quad (3)$$

The sub-task t_n specifies the information needed to advance the reasoning process, such as extracting a numerical quantity from a chart, identifying a constraint from the context, or retrieving a domain formula. The query set q_n operationalizes this sub-task into concrete information requests that can be answered by specialized agents. The termination flag s_n indicates whether the current evidence state is already

sufficient to answer the original question Q :

$$s_n = \begin{cases} \text{stop}, & \text{if } \mathcal{A}_P \text{ judges that } (E_{n-1}, R_{n-1}) \text{ is sufficient for } Q, \\ \text{continue}, & \text{otherwise.} \end{cases} \quad (4)$$

This formulation makes the planner adaptive rather than static. Instead of committing to a fixed chain of sub-tasks in advance, the planner revises subsequent reasoning goals according to the evidence accumulated so far. Such adaptive decomposition is particularly important in financial reasoning, where relevant information is often distributed across text, tables, charts, and external knowledge. The complete planner-agent prompt design is provided in Appendix B.1.

3.4. Multi-source evidence acquisition

Given the query set $q_n = \{q_{n,1}, \dots, q_{n,L_n}\}$ generated at step n , each query is routed to the most appropriate evidence-seeking agent:

$$a_{n,\ell} = \rho(q_{n,\ell}), \quad a_{n,\ell} \in \{\mathcal{A}_I, \mathcal{A}_C, \mathcal{A}_K\}, \quad (5)$$

where $\rho(\cdot)$ denotes the routing function. The corresponding evidence is then retrieved as

$$e_{n,\ell} = \begin{cases} \mathcal{A}_I(I, q_{n,\ell}), & \text{if } a_{n,\ell} = \mathcal{A}_I, \\ \mathcal{A}_C(C, q_{n,\ell}), & \text{if } a_{n,\ell} = \mathcal{A}_C, \\ \mathcal{A}_K(D, q_{n,\ell}), & \text{if } a_{n,\ell} = \mathcal{A}_K, \end{cases} \quad (6)$$

where D denotes the external financial knowledge base. The evidence returned at step n is collected into

$$e_n = \{e_{n,1}, \dots, e_{n,L_n}\}, \quad (7)$$

and the evidence state is updated by

$$E_n = E_{n-1} \cup e_n. \quad (8)$$

The three evidence-seeking agents serve complementary roles. **Image agent** $\mathcal{A}_I$ extracts structured visual evidence such as values, trends, and relations from charts, tables, and figures. **Context agent** $\mathcal{A}_C$ identifies task-relevant information from the textual context, including variables, assumptions, and problem constraints. **Knowledge agent** $\mathcal{A}_K$ retrieves financial concepts, formulas, and domain-specific principles from the external knowledge base D .

Table 2

Statistics of the constructed financial knowledge database D , including knowledge types and topic diversity.

	Number	Ratio
Database Size	8440	–
Knowledge Type		
Financial Concept	5698	67.51%
Math Formula	2742	32.49%
Topic Diversity		
Investment	1063	12.60%
Quantitative Methods	819	9.70%
Valuation and Risk Models	785	9.30%
Financial Markets and Products	776	9.20%
Financial Reporting and Analysis	701	8.30%
Portfolio Management	684	8.10%
Fixed Income	658	7.80%
Credit Risk	456	5.40%
Foundation of Risk Management	447	5.30%
Economics	414	4.90%
Operational Risk	346	4.10%
Derivatives	338	4.00%
Market Risk	329	3.90%
Corporate Finance	329	3.90%
Liquidity and Treasury Risk	295	3.50%

A key property of SEA is that evidence acquisition is *step-conditioned*. The system does not passively summarize all available inputs at once. Rather, it retrieves evidence only when demanded by the current sub-task. This improves both the relevance and the precision of intermediate evidence, which is essential for reliable multi-step financial reasoning. We provide the complete prompts used by the three evidence-seeking agents in Appendix B.2. These prompts specify how the Image Agent, Context Agent, and Knowledge Agent respond to planner-generated queries by extracting visual evidence from I , textual evidence from C , and financial knowledge from D , respectively.

Financial knowledge base. To alleviate the domain-knowledge limitations of general-purpose MLLMs, we construct an external financial knowledge base D from the training sets of FinMR and FAMMA. Specifically, we extract answer explanations, which often contain financial concepts, formulas, and reasoning procedures, and use an LLM to identify key terms together with their definitions and mathematical expressions across 15 financial topics. The resulting database provides structured domain knowledge that supports downstream evidence retrieval. The statistics of D are reported in Table 2.

3.5. Evidence-guided reasoning

After updating the evidence state, the reasoning agent $\mathcal{A}_R$ performs the current inference step. At step n , it takes as input the sub-task t_n , the updated evidence state E_n , and the previous reasoning trajectory R_{n-1} , and produces an intermediate reasoning output r_n :

$$r_n = \mathcal{A}_R(t_n, E_n, R_{n-1}). \quad (9)$$

The reasoning trajectory is then updated as

$$R_n = R_{n-1} \cup \{r_n\}. \quad (10)$$

The interaction between evidence acquisition and reasoning is tightly coupled. If the currently accumulated evidence is insufficient (i.e., the reasoning agent reports issues as a reasoning step), the planner generates a new sub-task and additional queries at the next step. If the planner determines that the evidence is sufficient, the iterative process terminates and the final answer is produced:

$$\hat{Y} = \mathcal{A}_P^{\text{final}}(X, E_N, R_N), \quad (11)$$

where N is the final reasoning step satisfying $s_N = \text{stop}$.

This design ensures that the final decision is grounded in an explicit evidence trail rather than implicit latent reasoning alone. Each reasoning step is therefore traceable to retrieved evidence, improving robustness and interpretability. The implementation details of the reasoning agent are provided in Appendix B.3.

3.6. Query-based communication protocol

SEA uses an effective query-based communication protocol to coordinate agents. Instead of relying on complex symbolic message passing or open-ended multi-agent debate, the planner communicates with evidence-seeking agents through concise natural-language queries. This design has two advantages. First, it aligns naturally with the capabilities of modern LLM/MLLM agents. Second, it constrains information exchange to the evidence required by the current sub-task, reducing redundancy and improving alignment between planning and retrieval.

The protocol operates as follows. At each step, the planner first decides whether reasoning should continue. If $s_n = \text{stop}$, the system terminates evidence acquisition and produces the final answer using the accumulated evidence and reasoning history. Otherwise, the planner generates a query set q_n , each query is routed to a specialized evidence-seeking agent, and the returned evidence is added to the shared memory for downstream reasoning. Algorithm 1 summarizes the complete SEA procedure.

Algorithm 1 SEA: step-wise evidence acquisition for multimodal financial reasoning.

Require: Input instance $X = (C, I, Q, O)$, knowledge base D , maximum number of steps $N_{\max}$

Ensure: Predicted answer $\hat{Y}$

```

1: Initialize evidence state  $E_0 \leftarrow \emptyset$ 
2: Initialize reasoning trajectory  $R_0 \leftarrow \emptyset$ 
3: for  $n = 1$  to  $N_{\max}$  do
4:    $(t_n, q_n, s_n) \leftarrow \mathcal{A}_P(X, E_{n-1}, R_{n-1})$ 
5:   if  $s_n = \text{stop}$  then
6:      $\hat{Y} \leftarrow \mathcal{A}_P^{\text{final}}(X, E_{n-1}, R_{n-1})$ 
7:     return  $\hat{Y}$ 
8:   end if
9:    $e_n \leftarrow \emptyset$ 
10:  for each query  $q_{n,\ell} \in q_n$  do
11:     $a_{n,\ell} \leftarrow \rho(q_{n,\ell})$ 
12:    if  $a_{n,\ell} = \mathcal{A}_I$  then
13:       $e_{n,\ell} \leftarrow \mathcal{A}_I(I, q_{n,\ell})$ 
14:    else if  $a_{n,\ell} = \mathcal{A}_C$  then
15:       $e_{n,\ell} \leftarrow \mathcal{A}_C(C, q_{n,\ell})$ 
16:    else if  $a_{n,\ell} = \mathcal{A}_K$  then
17:       $e_{n,\ell} \leftarrow \mathcal{A}_K(D, q_{n,\ell})$ 
18:    end if
19:     $e_n \leftarrow e_n \cup \{e_{n,\ell}\}$ 
20:  end for
21:   $E_n \leftarrow E_{n-1} \cup e_n$ 
22:   $r_n \leftarrow \mathcal{A}_R(t_n, E_n, R_{n-1})$ 
23:   $R_n \leftarrow R_{n-1} \cup \{r_n\}$ 
24: end for
25:  $\hat{Y} \leftarrow \mathcal{A}_P^{\text{final}}(X, E_{N_{\max}}, R_{N_{\max}})$ 
26: return  $\hat{Y}$ 

```

4. Experiment

4.1. Experimental setup

Our framework instantiates agents based on three frontier models: GPT-4o [1], Gemini-2.5-Flash [2], and Claude-4.5-Sonnet [3]. For each

Table 3

Statistics and characteristics of the evaluation datasets, including test set size, task-type distribution, main image types, knowledge level, and reasoning requirements.

Dataset	Test Size	Math	Expertise	Main Image Type	Knowledge Level	Knowledge Intensive?	Numerical Math?	Advanced Math?
FinMR	640	242 (37.8%)	398 (62.2%)	Charts, tables, diagrams, time series, maps	College	Yes	Yes	Yes
FAMMA	200	100 (50.0%)	100 (50.0%)	Charts, tables, diagrams	College	Yes	Yes	Yes
MMMU-Finance	1100	1100 (100.0%)	–	Tables	College	Yes	Yes	No

experiment, all agents are instantiated using the same backbone model to ensure a fair comparison across different model families. To improve reproducibility and reduce randomness in multi-step reasoning, we use deterministic decoding for all agents, with the temperature set to 0.0. Each agent is prompted to return structured JSON outputs, including the planner's stopping decision, subtask, target agent, and evidence-seeking queries. The full prompt templates for all agents are provided in Appendix B.

To address the domain knowledge limitations of general-purpose MLLMs, we further construct an external financial knowledge base D using expert-annotated explanatory data. See more details of the financial knowledge base in Section 3.4. These human-annotated resources provide detailed explanations of financial concepts, formulas, and application scenarios, and therefore serve as a high-quality repository of external financial expertise. The knowledge agent retrieves relevant financial concepts, formulas, and domain-specific rules from this database according to the query generated by the planner.

To keep the overall reasoning process computationally tractable and to avoid unnecessarily long planning chains, we set the maximum number of planning steps to $N_{\max} = 6$. At each step, the planner generates at most three evidence-seeking queries, corresponding to the image agent, context agent, and knowledge agent. Therefore, the maximum number of agent invocations for one sample is bounded by $1 + N_{\max}(1 + 3 + 1)$, where the three terms inside the parentheses correspond to one planner call, up to three evidence-agent calls, and one reasoning-agent call at each step, and the additional call denotes the final answer generation. In practice, the reasoning process may terminate earlier once the planner determines that the accumulated evidence is sufficient.

4.2. Benchmarks

We evaluate the proposed method on three financial multimodal reasoning benchmarks: FinMR, FAMMA, and MMMU-Finance. These benchmarks are selected because they differ in domain scope, question type, evidence dependency, and difficulty level, thereby allowing us to evaluate whether SEA generalizes across different financial reasoning scenarios. To prevent knowledge leakage, all experiments are conducted on the test sets only, since the training sets of FinMR and FAMMA are used to construct the external financial knowledge base. Table 3 summarizes the statistics and characteristics of the three testing datasets.

FinMR [9] is an expert-level financial multimodal reasoning benchmark consisting of 3200 financial reasoning questions. Each instance includes detailed textual context, one or more financial images, candidate options, and expert-annotated explanations. FinMR is designed to evaluate professional-level financial reasoning, covering diverse financial topics and requiring the integration of mathematical reasoning, domain knowledge, and visual interpretation. In our evaluation subset, FinMR contains both mathematical reasoning questions and financial expertise questions. The mathematical subset mainly requires numerical calculation, formula application, and quantitative comparison, while the expertise subset focuses more on conceptual understanding, financial interpretation, and domain-specific judgment. This makes FinMR the most knowledge-intensive and expert-oriented benchmark in our evaluation.

FAMMA [7] is a multilingual financial multimodal reasoning benchmark designed to assess college-level and advanced financial expertise. The original benchmark contains financial question-answer pairs collected from textbooks and exams, spanning multiple finance subfields such as corporate finance, asset management, and financial engineering. Each question is paired with one or more images, such as tables, charts, and diagrams, requiring models to combine textual understanding, visual evidence extraction, and financial knowledge. Compared with FinMR, FAMMA emphasizes multilinguality and academic financial knowledge. In our evaluation subset, FAMMA is evenly divided into mathematical reasoning and expertise-oriented questions, allowing us to test whether SEA is effective across both quantitative and conceptual financial reasoning.

MMMU-Finance [17] is a finance-focused subset of the Multi-discipline Multimodal Understanding benchmark. MMMU is designed to evaluate college-level multimodal understanding and deliberate reasoning across a wide range of academic disciplines. We use the finance-related subset, which contains samples drawn from Finance, Economics, and Accounting¹. Unlike FinMR and FAMMA, which are specifically constructed for financial multimodal reasoning, MMMU-Finance provides a broader academic setting with more standardized exam-style questions. Notably, approximately 95% of the visual inputs in this subset are financial tables, making it especially useful for evaluating table-based numerical reasoning and structured financial information extraction.

Overall, the three benchmarks provide complementary evaluation settings. FinMR tests expert-level financial reasoning with rich textual and visual evidence; FAMMA evaluates multilingual and college-level financial reasoning across heterogeneous visual inputs; and MMMU-Finance examines broader academic finance reasoning, with a strong emphasis on table-based visual evidence. Therefore, evaluating SEA on these datasets allows us to assess its generalization across different levels of difficulty, evidence sources, and financial reasoning types.

4.3. Baselines

The following baseline methods are compared in our experiments: **CoT** [44] is a method that instructs LLMs to generate intermediate reasoning steps (i.e., *Let's think step by step*) before producing the final answer. It serves as a strong single-agent baseline by improving reasoning transparency and decomposing complex financial problems into sequential inference steps.

MAD [45] studies the impact of decision-making protocols in multi-agent debate systems. Multiple agents first generate candidate solutions and exchange reasoning through several discussion rounds. A predefined decision protocol, such as voting or consensus, is then applied to determine the final answer.

RAA [18] is a role-aware multi-agent framework for financial question answering, comprising an Evidence Retriever, a CoT-based Generator, and an Expert Reviewer. The retriever gathers financial knowledge, the generator produces an initial answer, and the reviewer refines it through critique and correction.

¹ Economics and Accounting are closely related subfields within the broader financial domain and frequently appear in financial reasoning tasks.

Table 4

Performance comparison across three financial multimodal reasoning benchmarks: FinMR, FAMMA, and MMMU-Finance. Results are reported as accuracy (%). Math and Expertise denote subsets requiring quantitative reasoning and domain-specific financial knowledge, respectively. Numbers in parentheses indicate the absolute improvement over the CoT baseline. Bold indicates the best result in each column.

Model	Method	FinMR(Acc%)			FAMMA(Acc%)			MMMU-Finance(Acc%)
		ALL	Math(37.8%)	Expertise(62.2%)	ALL	Math(50%)	Expertise(50%)	Math
GPT-4o	CoT	54.17	49.40	57.07	54.50	53.00	56.00	62.77
	MAD	58.31 (↑4.14)	54.39 (↑4.99)	60.70 (↑3.63)	58.5 (↑4.00)	58.00 (↑5.00)	59.00 (↑3.00)	67.00 (↑4.23)
	RAA	59.17 (↑5.00)	53.21 (↑3.81)	62.79 (↑5.72)	60.00 (↑5.50)	59.00 (↑6.00)	61.00 (↑5.00)	69.00 (↑6.23)
	SEA(ours)	62.37 (↑8.23)	56.62 (↑7.22)	65.85 (↑8.78)	63.00 (↑8.50)	61.00 (↑8.00)	65.00 (↑9.00)	74.00 (↑11.23)
Gemini-2.5-Flash	CoT	61.20	56.79	63.88	61	58.00	64.00	63.77
	MAD	66.19 (↑4.99)	59.79 (↑3.11)	70.01 (↑6.13)	63.00 (↑2.00)	59.00 (↑1.00)	67.00 (↑3.00)	69.44 (↑5.67)
	RAA	69.42 (↑7.22)	62.44 (↑5.65)	72.06 (↑8.18)	64.50 (↑3.50)	60.00 (↑2.00)	62.00 (↑5.00)	71.23 (↑7.46)
	SEA(ours)	74.80 (↑13.60)	66.20 (↑9.41)	80.00 (↑16.12)	69.10 (↑8.10)	65.00 (↑7.00)	73.20 (↑9.20)	79.00 (↑15.23)
Claude-4-5-Sonnet	CoT	60.89	58.59	62.28	60.00	59.00	61.00	63.78
	MAD	74.46 (↑13.58)	72.31 (↑13.72)	75.77 (↑13.49)	61.00 (↑1.00)	60.00 (↑1.00)	62.00 (↑1.00)	73.55 (↑9.77)
	RAA	71.49 (↑10.61)	70.11 (↑11.52)	72.33 (↑10.05)	61.00 (↑1.00)	59.00 (-)	63.00 (↑2.00)	78.00 (↑14.22)
	SEA(ours)	78.49 (↑17.61)	76.50 (↑17.91)	79.70 (↑17.42)	64.50 (↑4.5)	63.00 (↑4.00)	66.00 (↑5.00)	82.70 (↑18.92)

5. Results

In this section, we evaluate SEA from multiple perspectives to understand both its overall effectiveness and the sources of its performance gains. We first compare SEA with representative baselines on three financial multimodal reasoning benchmarks (Section 5.1), and then provide a series of analyses to examine the contribution of its key design choices (Section 5.2), dynamic sub-task generation (Section 5.3), evidence acquisition quality (Section 5.4), agent communication (Section 5.5), and remaining error patterns (Section 5.6).

5.1. Overall results

Table 4 reports the performance comparison of different reasoning strategies across three multimodal financial reasoning benchmarks: FinMR, FinMMA, and MMMU-Finance. Overall, our proposed SEA method consistently achieves the best performance across all models and datasets, demonstrating the effectiveness of demand-driven evidence acquisition for multimodal financial reasoning.

Across all benchmarks, SEA significantly outperforms the standard Chain-of-Thought (CoT) baseline. For example, on the FinMR benchmark, SEA improves the overall accuracy from 54.17% to 62.37% for GPT-4o, from 61.20% to 74.80% for Gemini-2.5-Flash, and from 60.89% to 78.49% for Claude-4-5-Sonnet. Notably, the improvement for Claude-4-5-Sonnet exceeds 17 percentage points, indicating that the proposed framework can substantially enhance the reasoning capability of strong base models.

Compared with existing multi-agent reasoning methods such as MAD and RAA, SEA achieves consistently higher performance across all benchmarks. While MAD relies on multi-agent debate and RAA introduces role-aware reasoning agents, both approaches still process information from a static input context. In contrast, SEA explicitly models *evidence acquisition* as a *decision process*, enabling the system to dynamically retrieve task-relevant information from different sources before performing reasoning. This design leads to more reliable intermediate reasoning steps and reduces error propagation during multi-step reasoning.

The improvement of SEA is particularly pronounced on the mathematical problems of the three benchmarks, which require precise numerical reasoning and formula application. For example, on the FinMR Math subset, SEA achieves 66.20% accuracy with Gemini-2.5-Flash and 76.50% with Claude-4-5-Sonnet, substantially outperforming other reasoning strategies. This suggests that dynamically retrieving relevant formulas and financial concepts helps mitigate calculation errors and improves reasoning reliability.

Another important observation is that SEA consistently improves performance across all evaluated base models, including GPT-4o, Gemini-2.5-Flash, and Claude-4-5-Sonnet. Despite the large differences in model architectures and capabilities, the proposed framework yields stable gains across FinMR, FinMMA, and MMMU. This result indicates that SEA is model-agnostic and can serve as a general framework for enhancing multimodal reasoning in financial tasks.

To further assess the reliability of the observed improvements, we conduct a non-parametric statistical significance analysis based on the accuracy scores reported in Table 4. Since the compared methods are evaluated under the same model–dataset–subset setting and the normality assumption may not hold for accuracy scores, we use non-parametric tests in Table 5. Specifically, each model–dataset–subset combination is treated as one evaluation setting, resulting in 21 paired observations for each method. We first apply the Friedman test to compare whether there exist overall performance differences among CoT, MAD, RAA, and SEA. The Friedman test indicates a significant difference among the compared methods, $\chi^2_F(3) = 55.23$, $p = 6.14 \times 10^{-12}$, with the significance level $\alpha = 0.05$.

We further compare the average ranks of the methods. SEA obtains the best average rank of 1.00, followed by RAA (2.38), MAD (2.69), and CoT (3.93). The Nemenyi post-hoc test shows that the rank differences between SEA and the baseline methods are larger than the critical difference at $\alpha = 0.05$, indicating that SEA performs significantly better than the compared baselines across evaluation settings. In addition, we perform two-sided Wilcoxon signed-rank tests to further confirm the improvements of SEA over CoT ($p = 9.54 \times 10^{-7}$); MAD ($p = 5.94 \times 10^{-5}$); and RAA ($p = 5.93 \times 10^{-5}$). These results suggest that the performance gains of SEA are statistically significant under the model–dataset–subset setting rather than being driven by isolated cases.

Overall, the results demonstrate that explicitly separating *information acquisition* from *reasoning* is crucial for solving complex multimodal financial problems. By dynamically retrieving task-relevant evidence before reasoning, SEA significantly improves both reasoning accuracy and robustness compared with existing prompting-based and multi-agent approaches.

5.2. Ablation study

To quantify the contribution of each component in SEA, we conduct ablation studies by systematically removing key modules from the full framework. Since the full SEA system built on Claude-4-5-Sonnet achieves the strongest overall performance in Table 4, we use this model as the base for all ablation experiments. These analyses are intended to examine how visual grounding, multi-agent communication, and spe-

Table 5

Statistical significance analysis across 21 model–dataset–subset evaluation settings. The significance level is set to $\alpha = 0.05$.

Statistical Test	Comparison	Result
Friedman test	CoT vs. MAD vs. RAA vs. SEA	$\chi^2_F(3) = 55.23, p = 6.14 \times 10^{-12}$
Friedman average rank	SEA / RAA / MAD / CoT	1.00 / 2.38 / 2.69 / 3.93
Nemenyi post-hoc test	SEA vs. all baselines	Significant at $\alpha = 0.05$
Wilcoxon signed-rank test	SEA vs. CoT	$p = 9.54 \times 10^{-7}$
Wilcoxon signed-rank test	SEA vs. MAD	$p = 5.94 \times 10^{-5}$
Wilcoxon signed-rank test	SEA vs. RAA	$p = 5.93 \times 10^{-5}$

Table 6

Ablation study of SEA on FinMR, FAMMA, MMMU-Finance, using base model Claude-4.5-Sonnet.

Method	FinMR%	FAMMA%	MMMU%
Full SEA	78.49	64.50	82.70
w/o Planner	63.79	60.00	65.30
w/o Agent Communication	64.69	61.00	73.21
w/o Evidence Seeking Agents	72.19	62.30	78.22

cialized evidence acquisition jointly affect performance on financial multimodal reasoning.

SEA w/o Planner. In the full framework, the planning agent decomposes the task into step-wise sub-tasks and generates targeted queries to guide evidence acquisition at each reasoning step. In this variant, we remove the planning agent entirely. Instead, each evidence-seeking agent operates based on a fixed, generic instruction (i.e., *understanding the image, context, and question to extract relevant evidence, or retrieving knowledge from a financial knowledge base*), without step-specific guidance. The collected evidence is then directly used by the reasoning agent. As shown in Table 6, this variant achieves performance slightly better than standard CoT reasoning in Table 4, but remains significantly inferior to the full SEA framework. This observation highlights that while evidence-seeking agents are beneficial, their effectiveness critically depends on fine-grained, step-conditioned guidance. Without an explicit planning mechanism, evidence retrieval becomes coarse and less aligned with the evolving reasoning needs, leading to suboptimal downstream reasoning performance.

SEA w/o Agent Communication. In this variant, we remove the iterative communication mechanism among the planning agent, the evidence-seeking agents, and the reasoning agent. The planner generates the reasoning plan and queries only once, the evidence-seeking agents retrieve information in a single pass, and the reasoning agent produces the final answer from the collected evidence. Compared with the full SEA system, this variant yields the largest overall degradation, dropping to 64.69% on FinMR, 61.00% on FAMMA, and 73.21% on MMMU-Finance. These results indicate that iterative multi-agent communication substantially improves the quality of intermediate evidence and enables more effective coordination between planning, evidence collection, and reasoning.

SEA w/o Evidence-Seeking Agents. To evaluate the importance of targeted evidence acquisition, we remove the evidence-seeking agents and retain only the planner and reasoning agent. Under this setting, the system can no longer extract evidence from different sources through specialized agents. Performance again declines across all datasets, reaching 72.19% on FinMR, 62.30% on FAMMA, and 78.22% on MMMU-Finance. Although this variant remains stronger than the settings without planner vision input or agent communication, it is still consistently worse than the full SEA system, showing that dedicated evidence-seeking agents contribute meaningfully to accurate financial multimodal reasoning.

Overall, the ablation results confirm that all three components are beneficial, while their contributions are not equally strong. Iterative agent communication produces the largest performance gain, suggesting that coordination among agents is central to SEA. Direct planner access to visual input is also important, particularly for tasks requiring precise visual grounding. Specialized evidence-seeking agents further improve performance by supporting targeted information acquisition from heterogeneous sources. The full SEA system, therefore, achieves the best results on FinMR, FAMMA, and MMMU-Finance, demonstrating the effectiveness of the proposed multi-agent reasoning framework.

5.3. Utility analysis of dynamic sub-task generation

We compare dynamic sub-task generation with a static decomposition strategy on the FinMR dataset, where all sub-tasks are predefined at the beginning and remain fixed throughout the reasoning process. As illustrated in Fig. 3, dynamic sub-task generation consistently outperforms static decomposition across all base models. In particular, the performance gains are substantial for Gemini-2.5-Flash (+9.43%) and noticeable for GPT-4o (+3.93%), indicating that the benefit of dynamic planning is robust across models of varying capabilities.

The performance improvement stems from the adaptive nature of dynamic planning. In static decomposition, the reasoning trajectory is determined without access to intermediate evidence, making it vulnerable to early-stage planning errors that cannot be corrected later. This limitation is especially pronounced in FinMR, where relevant information is often fragmented across textual context, visual inputs, and domain-specific knowledge, requiring flexible and iterative reasoning. In contrast, dynamic sub-task generation enables the planner to iteratively update subsequent sub-tasks conditioned on the evolving evidence state. This allows the system to refine reasoning objectives based on newly acquired evidence, correct earlier misinterpretations, and focus on the most relevant information at each step. As reflected in Fig. 3, this adaptive process leads to consistent and sometimes large performance improvements across all evaluated models.

5.4. Utility analysis of evidence acquisition

To evaluate the quality of evidence acquired by different agents, we conduct a fine-grained analysis on the FinMR, FAMMA, and MMMU-Finance benchmarks.

Case-based Analysis. For each dataset, we randomly sample 100 intermediate queries generated during the reasoning process and assess the correctness of the corresponding agent responses. We adopt different evaluation strategies tailored to each modality. For the image agent, correctness is assessed via human evaluation due to the complexity of visual interpretation. For the context agent, we use GPT-5 to judge whether the generated answers are correct given the query and context, following prior work on LLM-based evaluation [46,47]. For the knowledge agent, we compare its responses with ground-truth explanations provided in the datasets, and employ an LLM to determine correctness.

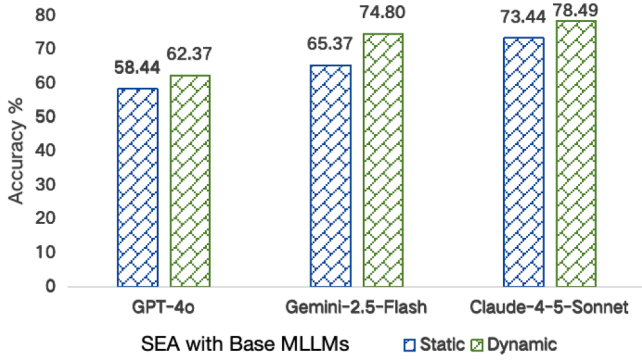


Fig. 3. Static vs. dynamic task decomposition on the FinMR benchmark. Dynamic sub-task generation consistently outperforms static planning across all models, highlighting the effectiveness of iterative, evidence-conditioned task refinement.

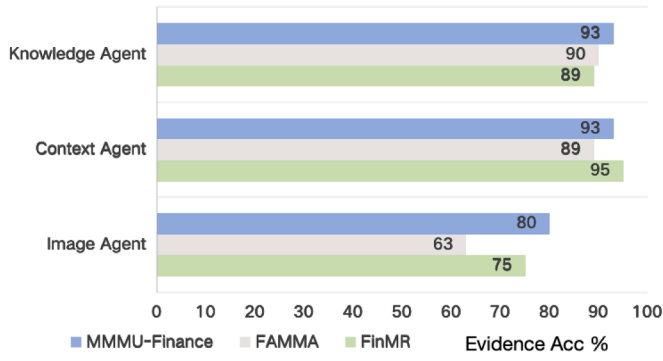


Fig. 4. Evidence quality of different agents on FinMR, FAMMA, and MMMU-Finance. Context and knowledge agents achieve consistently high accuracy, while the image agent shows relatively lower performance due to the challenges of fine-grained visual understanding.

Fig. 4 indicates that all three agents produce high-quality evidence overall, while exhibiting distinct characteristics across modalities. The context agent achieves consistently strong performance across datasets (e.g., 95% on FinMR), indicating its reliability in extracting and reasoning over textual context. The knowledge agent also performs robustly (up to 93% on MMMU-Finance), demonstrating its effectiveness in retrieving and applying domain-specific financial knowledge. In contrast, the image agent shows comparatively lower accuracy (e.g., 63% on FAMMA), reflecting the inherent difficulty of fine-grained visual perception in financial charts and tables. Nevertheless, its performance remains reasonably strong (75% on FinMR), suggesting that it still provides useful complementary evidence, particularly when visual information is essential.

The differences across agents highlight an important insight: evidence quality is modality-dependent, and errors in multimodal reasoning often originate from weaker perception modules. This observation further justifies the design of SEA, where multiple specialized agents collaboratively contribute evidence, allowing stronger modalities (e.g., the context and knowledge agents) to compensate for weaker ones (e.g., the image agent). Overall, these results demonstrate that SEA can reliably acquire high-quality intermediate evidence across modalities, which serves as a critical foundation for improving downstream reasoning performance.

Evidence Perturbation Analysis. To further examine how intermediate evidence quality affects final reasoning outcomes, we conduct an evidence perturbation analysis using Claude-4-5-Sonnet as the backbone model. For each benchmark, we independently drop knowledge evi-

dence, image evidence, and textual-context evidence at ratios of 0%, 25%, 50%, 75%, and 100%, and then evaluate the resulting final reasoning accuracy. This controlled setting allows us to quantify how the completeness of different evidence types affects the final answer. To provide a more statistically reliable assessment, we further report 95% Wilson confidence intervals for each accuracy value, which are visualized as error bars in Fig. 5.

As shown in Fig. 5, accuracy generally decreases as the evidence drop ratio increases across all three benchmarks, indicating that intermediate evidence directly supports the final reasoning process. On FinMR, the accuracy drops from 78.49% to 74.23%, 73.86%, and 73.24% when knowledge, textual-context, and image evidence are fully removed, respectively. The largest degradation comes from removing image evidence, suggesting that visual grounding is particularly important for FinMR. On FAMMA, the performance decreases from 64.50% to 63.25%, 63.08%, and 63.40% under full removal of knowledge, image, and textual-context evidence, respectively. Although the absolute drops are smaller, the consistent downward trend shows that all three evidence types contribute to the final prediction. On MMMU-Finance, accuracy decreases from 82.70% to 79.58%, 79.33%, and 79.75% when knowledge, image, and textual-context evidence are fully removed, respectively, again showing that both multimodal evidence and financial knowledge are important for reliable reasoning.

Overall, these perturbation results, together with the reported confidence intervals, provide direct empirical evidence that SEA's effectiveness is closely tied to the completeness of intermediate evidence. The performance degradation under increasing drop ratios confirms the utility and quality of the intermediate evidence: if the acquired evidence were mostly noisy, redundant, or weakly related to the reasoning task, removing a larger portion of it would not lead to a consistent decline in performance. Instead, the observed degradation suggests that SEA collects evidence that contributes positively to reasoning by constraining candidate answers, and supporting intermediate decisions. Last but not least, since only one evidence type is removed at a time, the remaining evidence sources can still partially compensate for the missing modality, highlighting the complementary nature of SEA's multi-agent evidence acquisition mechanism.

5.5. Utility analysis of query-based agent communication

To evaluate the effectiveness of different collaboration mechanisms, we compare four representative paradigms: 1) pipeline-based collaboration, where agents operate sequentially without interaction; 2) controller-based coordination, where a central planner assigns subtasks to different modules without explicit query-response communication; 3) debate-based collaboration, where agents engage in multi-round adversarial discussions; and 4) our query-driven interactive collaboration, which enables structured query-response communication between the planner and evidence-seeking agents. For a fair comparison, all communication baselines are implemented using the same backbone model, the same input context. The pipeline-based baseline follows a sequential evidence-retrieval-and-reasoning process, similar to role-aware financial agents [18]. The controller-based baseline follows the centralized coordination paradigm of AutoGen-style systems [19], where a controller assigns subtasks to specialized agents but does not use explicit query-response evidence acquisition. The debate-based baseline follows multi-agent debate protocols [20], where multiple agents independently generate answers and exchange critiques before producing a final decision. In contrast, SEA uses planner-generated, step-conditioned queries to explicitly acquire missing evidence from the most relevant agent at each reasoning step.

As shown in Table 7, query-driven interaction consistently outperforms all other collaboration paradigms across all benchmarks. On FinMR, our method achieves 78.49% accuracy, significantly surpassing pipeline-based (+8.18%), controller-based (+11.05%), and debate-based (+6.16%) collaboration. Similar improvements are observed on

Table 7

Comparison with representative multi-agent communication paradigms. All methods are implemented using the same backbone model and input setting for fair comparison. The results show that the proposed query-based communication paradigm consistently outperforms pipeline-based, controller-based, and debate-based alternatives across all three benchmarks.

Communication Method	FinMR	FAMMA	MMMU-Finance
Pipeline-based: Role-aware Agent [18]	70.31	60.33	67.66
Controller-based: AutoGen-style [19]	67.44	56.78	65.59
Debate-based: Multi-agent Debate [20]	72.33	57.32	70.69
Query-based: SEA (ours)	78.49	64.50	82.70

FAMMA and MMMU-Finance, where query-driven interaction yields the best performance, with particularly notable gains on MMMU-Finance (+17.11% over controller-based and +15.04% over pipeline-based).

The results reveal several important insights. First, purely sequential collaboration is insufficient for complex multimodal financial reasoning. Pipeline-based methods lack the ability to dynamically acquire missing information, leading to incomplete or incorrect intermediate evidence. This limitation is especially pronounced in tasks requiring cross-modal integration and domain knowledge. Second, controller-based coordination, despite introducing task routing, remains limited in its ability to support evidence-grounded reasoning. Although the planner can decide which module to invoke, it delegates entire subtasks rather than explicitly identifying and retrieving missing information. As a result, the reasoning process remains implicit, and the system may fail to obtain the precise evidence required for accurate decision-making. Third, while debate-based collaboration introduces stronger interaction, unstructured or adversarial communication does not necessarily improve reasoning performance. The relatively lower performance on FAMMA suggests that multi-round discussions may introduce redundancy or distract from the core task, without effectively addressing missing evidence. Moreover, debate-based methods incur higher communication overhead without guaranteeing more accurate information acquisition.

In contrast, our query-driven framework achieves the best performance by enabling targeted and efficient evidence acquisition. At each step, the planner explicitly identifies missing information and formulates task-specific queries to retrieve evidence from the most relevant agent. This structured communication paradigm avoids both the rigidity of pipeline-based execution and the inefficiency of open-ended debate, while overcoming the implicit reasoning limitations of controller-based coordination. As a result, it leads to more accurate intermediate evidence and, consequently, more reliable final predictions. These re-

sults demonstrate that not all forms of interaction are equally effective. Instead of no interaction (pipeline), implicit task delegation (controller-based), or excessive interaction (debate), minimal yet targeted query-based communication provides the most suitable balance between efficiency and reasoning accuracy for multimodal financial tasks.

5.6. Error analysis

Although SEA achieves strong performance on FinMR and MMMU-Finance, we observe relatively lower accuracy on the FAMMA benchmark. To examine the limitations of our system on the benchmark, FAMMA, we conduct a detailed error analysis. Following prior works [48–50], we randomly sampled 100 error cases from the FAMMA dataset. Fig. 6 demonstrates the error distribution intuitively. The errors were categorized as follows:

Cross-Image Reasoning Errors (46%): The largest portion of errors occurs in one question with multiple charts or tables. These questions require the model to extract numerical information from several visual sources and aggregate them correctly. In many cases, the system fails to accurately combine information across different charts, leading to incorrect intermediate values and final answers.

Text-Image Alignment Errors (23%): Another common error arises when questions contain long textual contexts accompanied by images. In such cases, the model sometimes struggles to properly align textual descriptions with relevant visual elements. As a result, important information from either the text or the image may be overlooked during reasoning.

Scenario-Specific Knowledge Errors (10%): We also observe errors related to financial knowledge application. Although the system can retrieve general financial formulas or concepts, some questions require adapting these formulas to specific financial scenarios. In these cases, the model occasionally applies standard formulas without correctly accounting for contextual assumptions, leading to reasoning mistakes.

Others (21%): The remaining errors include numerical calculation mistakes and occasional reasoning inconsistencies.

Beyond the above empirical categories, these errors also reveal several component-level limitations of SEA. First, the framework depends on the quality of planner-generated subtasks. When a question requires integrating multiple charts, tables, or textual constraints, an incomplete subtask decomposition may cause the system to overlook necessary evidence. Second, although evidence-seeking agents improve modularity, they may still return incomplete, or partially misaligned evidence, especially when long textual contexts must be matched with specific

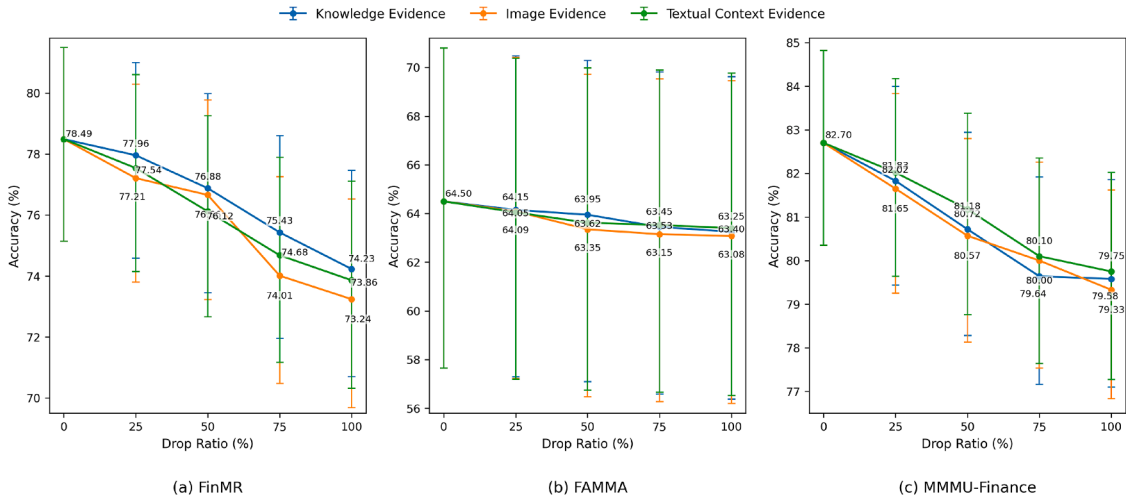


Fig. 5. Evidence perturbation analysis under different evidence drop ratios on three benchmarks. (a) FinMR, (b) FAMMA, and (c) MMMU-Finance. For each benchmark, we progressively drop knowledge, image, and context evidence from 0% to 100% and report the corresponding reasoning accuracy. Error bars indicate 95% Wilson confidence intervals.

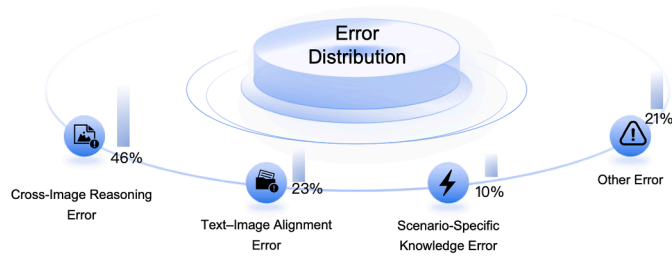


Fig. 6. Error distribution in FAMMA.

visual elements. Third, the current framework does not explicitly resolve conflicts among evidence returned by different agents. As a result, contradictory or insufficient evidence may propagate to the reasoning agent and lead to incorrect intermediate reasoning. Finally, although the Knowledge Agent can retrieve general financial concepts and formulas, some questions require adapting these principles to scenario-specific assumptions, which remains challenging. These limitations suggest that future work should incorporate planner verification, cross-source evidence consistency checking, and more robust mechanisms for aggregating evidence across multiple visual and textual sources.

6. Conclusion

In this work, we present SEA, a multi-agent framework for multimodal financial reasoning that explicitly models the interaction between task decomposition, evidence acquisition, and reasoning. By introducing a planner-guided, query-based communication protocol, SEA enables dynamic retrieval of step-conditioned evidence from heterogeneous sources, including visual, textual, and knowledge-based inputs. The key insight of SEA is that reasoning errors in multimodal financial tasks often stem from insufficient or incorrect intermediate evidence. By maintaining an explicit evidence state and grounding each reasoning step in retrieved information, SEA effectively improves interpretability. Empirical results demonstrate that our approach achieves consistent gains across multiple benchmarks, particularly in scenarios requiring cross-modal integration and domain-specific knowledge. Overall, SEA effectively integrates perception, knowledge, and reasoning in complex multimodal settings and offers a promising direction toward more reliable and interpretable financial reasoning systems.

7. Limitations

While SEA demonstrates strong performance improvements, several limitations remain. First, the framework relies on multiple large model-based agents, which introduces additional computational overhead compared to single-model approaches. Although the query-based communication protocol is simple, the iterative interaction among agents may increase latency in real-world applications. Second, the effectiveness of SEA depends on the quality of the evidence-seeking agents. Errors in visual perception, information extraction, or knowledge retrieval can still propagate through the system, potentially affecting downstream reasoning. Finally, our experiments focus on financial multimodal benchmarks, and the generalization of SEA to other domains, such as scientific or medical reasoning, remains to be further explored.

CRedit authorship contribution statement

Shuangyan Deng: Writing – original draft, Validation, Software, Methodology, Conceptualization; **Zihao Huang:** Validation, Investigation, Data curation; **Kelvin Du:** Writing – review & editing, Investigation; **Rui Mao:** Writing – review & editing, Visualization; **Jiamou Liu:** Writing – review & editing, Supervision; **Ciprian Doru Giurcăneanu:** Writing – review & editing, Supervision, Project administration; **Erik Cambria:** Writing – review & editing, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

Appendix A. Summary of notations

The main notations of this work can be viewed in Table A.8.

Table A.8

Summary of the main notations used in the SEA framework.

Notation	Description
X	A multimodal financial reasoning instance.
C	Textual context of the question.
$I = \{i_1, \dots, i_S\}$	Set of visual inputs, such as charts, tables, or figures.
Q	Financial reasoning question.
$O = \{o_1, \dots, o_M\}$	Set of candidate answer options.
Y^*	Ground-truth answer.
$\hat{Y}$	Predicted answer.
E_n	Evidence state after reasoning step n .
e_n	Evidence retrieved at step n .
$e_{n,\ell}$	Evidence returned for the ℓ -th query at step n .
R_n	Reasoning trajectory after step n .
r_n	Intermediate reasoning output at step n .
t_n	Sub-task generated by the planner at step n .
$q_n = \{q_{n,1}, \dots, q_{n,L_n}\}$	Set of evidence-seeking queries at step n .
s_n	Termination flag indicating whether to continue or stop.
$\mathcal{A}_P$	Planner agent.
$\mathcal{A}_I$	Image evidence-seeking agent.
$\mathcal{A}_C$	Context evidence-seeking agent.
$\mathcal{A}_K$	Knowledge evidence-seeking agent.
$\mathcal{A}_R$	Reasoning agent.
$\mathcal{D}$	External financial knowledge base.
$\rho(\cdot)$	Routing function that assigns each query to an evidence-seeking agent.
$N_{\max}$	Maximum number of reasoning/planning steps.

Appendix B. Prompt templates for SEA

To improve reproducibility, we provide the prompt templates used by the agents in SEA. Each agent is instantiated with the same backbone LLM within an experiment, but is assigned a role-specific instruction. The planner agent generates step-conditioned subtasks and evidence-seeking queries; the evidence-seeking agents return modality-specific evidence; and the reasoning agent performs evidence-grounded inference based on the accumulated evidence state.

B.1. Planner agent prompt

Planner Agent Prompt

[System Input]:

You are the planner agent in a step-wise evidence acquisition framework for multimodal financial reasoning. Given the original instance, the accumulated evidence, and the intermediate reasoning trajectory, your task is to decide whether the current evidence is sufficient to answer the question. If the evidence is sufficient, output **stop**. Otherwise, generate the next sub-task and formulate targeted evidence-seeking queries.

The available evidence-seeking agents are: **Image Agent**, which extracts information from charts, tables, figures, and images; **Context Agent**, which retrieves information from the textual context; and **Knowledge Agent**, which provides financial concepts, formulas, definitions, and domain knowledge.

You should generate only the next necessary sub-task, avoid redundant queries, and assign each query to exactly one agent. Do not answer the question directly. Stop only when the accumulated evidence and reasoning trajectory are sufficient for the reasoning agent to derive the final answer. The maximum number of planning steps is $N_{\max} = 6$.

Return only a JSON object in the following format:

```
{
  "status": "continue" or "stop",
  "subtask": "<next reasoning objective, or None>",
  "queries": [
    {
      "agent": "Image Agent / Context Agent / Knowledge Agent",
      "query": "<targeted evidence-seeking query>"
    }
  ]
}
```

If **status** is "stop", set **subtask** to None and **queries** to an empty list.

[User Input]:

Context: {context}

Question: {question}

Options: {options}

Images: {images}

Step ID: {step_id}

Accumulated Evidence: {evidence_state}

Intermediate Reasoning Trajectory: {reasoning_trajectory}

Return only the JSON object.

B.2. Evidence-seeking agent prompts

Image Agent Prompt

[System Input]:

You are the Image Agent, a step-wise evidence acquisition framework for multimodal financial reasoning. Given a planner query and the visual inputs, your role is to extract only the visual evidence required by the query. The visual inputs may include charts, tables, figures, plots, or screenshots.

You should identify relevant numerical values, labels, axes, units, table entries, trends, comparisons, and relations from the images. Do not infer unsupported information. If the requested evidence is not visible, state that it is not available.

Return only the following JSON object:

```
{
  "agent": "Image Agent",
  "query": "<planner query>",
  "evidence": "<extracted visual evidence>"
}
```

[User Input]:

Images: {images}

Planner Query: {query}

Return only the JSON object.

Context Agent Prompt

[System Input]:

You are the Context Agent in SEA, a step-wise evidence acquisition framework for multimodal financial reasoning. Given a planner query and the textual context, your role is to extract only the textual evidence required by the query.

You should identify relevant facts, variables, assumptions, constraints, numerical quantities, temporal conditions, definitions stated in the context, and relations among entities or variables. If the requested evidence is not stated or implied in the context, state that it is not available.

Return only the following JSON object:

```
{
  "agent": "Context Agent",
  "query": "<planner query>",
  "evidence": "<extracted textual evidence>"
}
```

[User Input]:

Context: {context}

Planner Query: {query}

Return only the JSON object.

Knowledge Agent Prompt

[System Input]:

You are the Knowledge Agent in SEA, a step-wise evidence acquisition framework for multimodal financial reasoning. Given a planner query and retrieved entries from the external financial knowledge base $\mathcal{D}$, your role is to provide only the financial knowledge required by the query. You should retrieve relevant financial concepts, mathematical formulas, definitions, and domain-specific reasoning principles. If the requested knowledge is not available in the retrieved entries, state that it is not available.

Return only the following JSON object:

```
{  
  "agent": "Knowledge Agent",  
  "query": "<planner query>",  
  "evidence": "<retrieved financial knowledge>"  
}
```

[User Input]:

Retrieved Knowledge Entries: {retrieved__knowledge__entries}

Planner Query: {query}

Return only the JSON object.

B.3. Reasoning agent prompt

Reasoning Agent Prompt

[System Input]:

You are the Reasoning Agent in SEA, a step-wise evidence acquisition framework for multi-modal financial reasoning.

At each reasoning step, you are given the current sub-task, the updated evidence state, and the previous reasoning trajectory. Your role is to perform only the current inference step based on the available evidence. You should produce an intermediate reasoning output that advances the reasoning trajectory.

You must follow these rules:

1. Use only the accumulated evidence provided in the evidence state and the previous reasoning trajectory.
2. Ground the current reasoning step explicitly in the retrieved evidence.
3. Do not introduce unsupported assumptions or external knowledge unless it is already included in the evidence state.
4. Do not answer the final question unless the current sub-task explicitly asks for final decision synthesis.
5. If the evidence is insufficient for the current sub-task, report the missing or unclear information as the reasoning output, so that the planner can generate additional evidence-seeking queries in the next step.
6. Keep the reasoning focused on the current sub-task rather than solving the entire problem at once.

Return only a JSON object in the following format:

```
{
  "subtask": "<current sub-task>",
  "evidence_used": [
    "<relevant evidence item 1>",
    "<relevant evidence item 2>"
  ],
  "reasoning_output": "<intermediate result for the current sub-task>",
  "evidence_issue": "<None, or a description of insufficient evidence>"
}
```

If the available evidence is sufficient for the current sub-task, set `evidence_issue` to `None`. If the available evidence is insufficient, explain what evidence is missing in `evidence_issue` and do not infer beyond the evidence.

[User Input]:

Current Sub-task: {subtask}

Updated Evidence State: {evidence_state}

Previous Reasoning Trajectory: {reasoning_trajectory}

Return only the JSON object.

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